Jamie Shih

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Education

University of California, Santa Barbara **Sep. 2024 – Jun. 2028**
B.S. Electrical Engineering, Minor in Music *Santa Barbara, CA*

Research Experience

Geometric Intelligence Lab, UCSB ECE **Sep. 2026 – Present**
Lab Manager – PI: Prof. Nina Miolane *Santa Barbara, CA*
- Develop agentic AI workflows and Claude skills to support research automation and computational experiments.
- Coordinate server updates and software maintenance to sustain research infrastructure and experiment uptime.

Music Cognition Lab, UCSB Music Department **Sep. 2026 – Present**
Research Assistant – PI: Dr. Janet Bourne *Santa Barbara, CA*
- Study music perception, categorization, and cross-modal associations through behavioral and corpus-based research.
- Apply R-based quantitative analysis to music cognition data; explore potential applications of audio signal processing.

Work Experience

California NanoSystems Institute (CNSI), UCSB **Jul. 2026 – Present**
Cleanroom Assistant – Quantum Structures Cleanroom Facility *Santa Barbara, CA*
- Maintain nanofabrication cleanroom operations; troubleshoot vacuum, chiller, and gas-delivery subsystems.
- Support technical projects across cleanroom tools, including calibration, deposition systems, and leak testing.

University of California, Santa Barbara **Aug. 2025 – Present**
Resident Assistant – Women in STEM Residential Community *Santa Barbara, CA*
- Serve as RA for a Women in STEM community in a 600+ resident hall, supporting women in science and engineering.
- Facilitate academic advising for STEM residents, connecting them with advisors, campus resources, and peer mentors.

Gateway Educational Services **Jan. 2026 – Sep. 2026**
STEM Tutor *Santa Barbara, CA*
- Tutored 3–5 students weekly in calculus and STEM subjects, adapting instruction to individual learning needs.
- Led STEM Robotics Camp sessions teaching Python, C++, Arduino, and foundational problem-solving skills.

Projects

Digital Design & FPGA Projects | *Verilog, FPGA, Digital Logic* **Mar. 2026 – Jun. 2026**
- Designed FPGA-based synchronous digital systems (Mealy/Moore FSMs) and wrote Verilog testbenches, including a simulated traffic-light controller, to verify logic and timing behavior.

Arduino Lightspeed Game | *Arduino, C++, Electronics, Soldering* **Jan. 2025 – Mar. 2025**
- Designed and assembled an interactive Arduino game using a 7-segment display, LED strip, and custom C++ code.
- Soldered and wired components; designed 3D-printed buttons and hardware for the physical interface.

Lights on Greenmoor – Light Show Design | *DMXIS, DMX Lighting, 3D CAD Simulation* **2022 – 2025**
- Programmed and ran a public light show in DMXIS, building DMX cues and sequences timed to music.
- Modeled fixtures and previewed looks in 3D CAD simulations, reducing on-site programming and setup time.

Technical Skills

Programming: Python, C++, Java, R, MATLAB, Verilog, HTML/CSS, Arduino
Hardware & EDA: Circuit Design, Digital Logic, FPGA, PCB Design, KiCad, LTspice, AutoCAD, Soldering
Lab & Cleanroom: Cleanroom Procedures, Equipment Maintenance, Troubleshooting, Safety Protocols
Tools: Git/GitHub, Claude, DMXIS, DMX Lighting, 3D CAD Simulation, DaVinci Resolve, Adobe, CapCut, Microsoft Office

Leadership & Activities

IEEE at UC Santa Barbara **Sep. 2025 – Present**
Member *Santa Barbara, CA*
- Build and solder small PCB projects, collaborate with peers on hardware projects, and attend technical workshops.

Girls Who Code **Sep. 2021 – Present**
Club Publicity & Tutoring Volunteer *Irvine, CA*
- Ran club publicity, creating outreach materials and social media content to recruit girls interested in computer science.
- Tutored elementary and middle school students in Python, Java, and programming fundamentals through hands-on projects.